# Chapter 4 — Use cases and applications

Abstract schemas become concrete in domain settings

# Learning objectives

- Describe annotation demands in autonomous driving and medical imaging
- Recognize temporal and spatial consistency needs in emerging geospatial and

# Case study — Autonomous vehicles

- Lane markings
- Labels feed perception and planning stacks
- Quality-control practices and geographic diversity in collection matter as much as volume

# Case study — Healthcare

- Medical imaging annotation marks lesions
- Expert clinicians typically validate or produce gold labels
- Privacy and consent constraints govern what can be labeled and shared
- Segmentation precision often matters more than coarse boxes because boundary errors can

# Case study — E-commerce

- Retail catalogs use category and attribute labels for search, recommendations
- Product taxonomies and review polarity labels must stay consistent across locales and
- A shoe labeled “sneaker” in one locale and “trainer” in another without a mapped taxonomy

# Emerging applications

- Augmented reality needs spatially registered object labels so virtual content aligns with
- Climate and remote-sensing workflows need geospatial segmentation over time series
- Both stress temporal consistency and domain-expert review beyond single-frame box labeling
- Emerging domains reuse the same modality techniques from earlier clips but raise the cost

# Takeaways

- Autonomous driving couples volume with safety-critical QC
- Healthcare pairs expert gold labels with privacy governance
- E-commerce needs stable taxonomies across locales
- Emerging AR and geospatial work add temporal consistency to spatial labels

# Next

- Complete the quiz for this part
- Pre-labeling as proposals rather than shortcuts